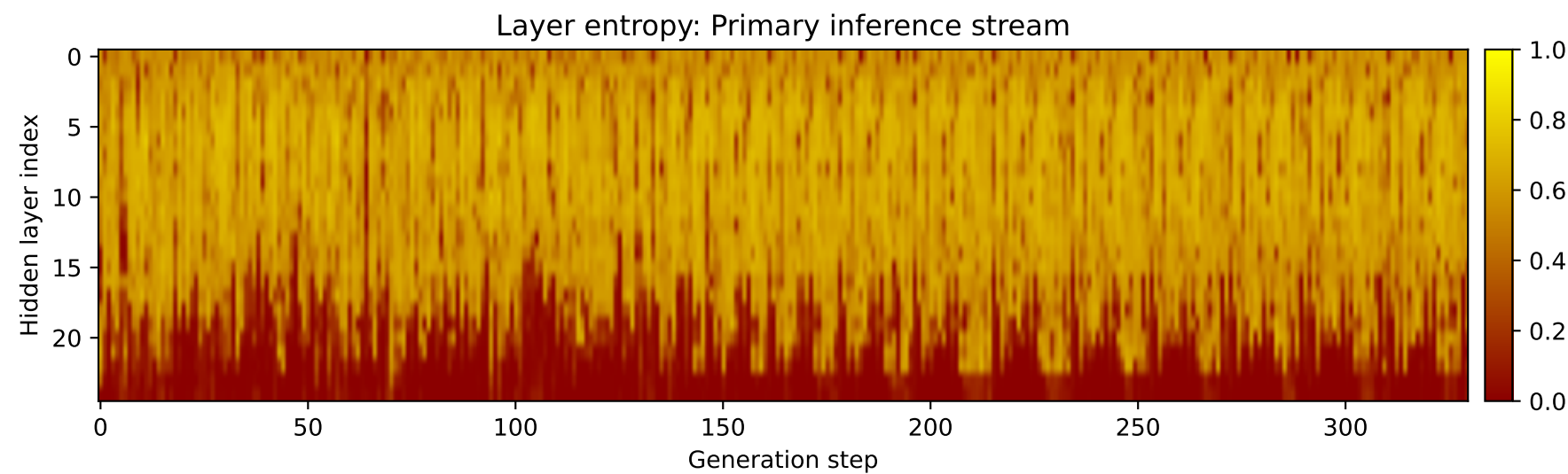
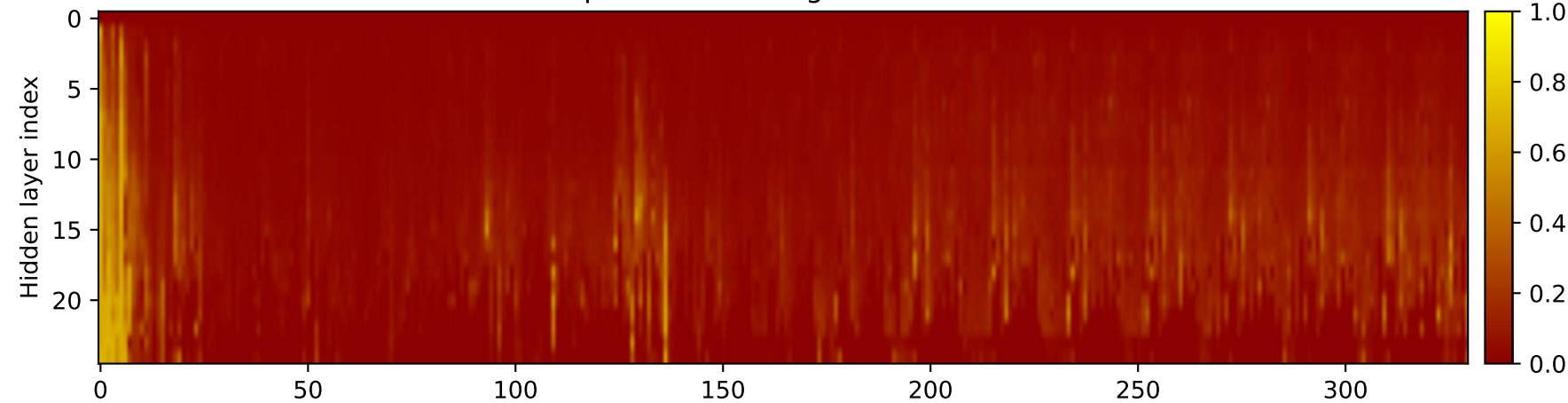
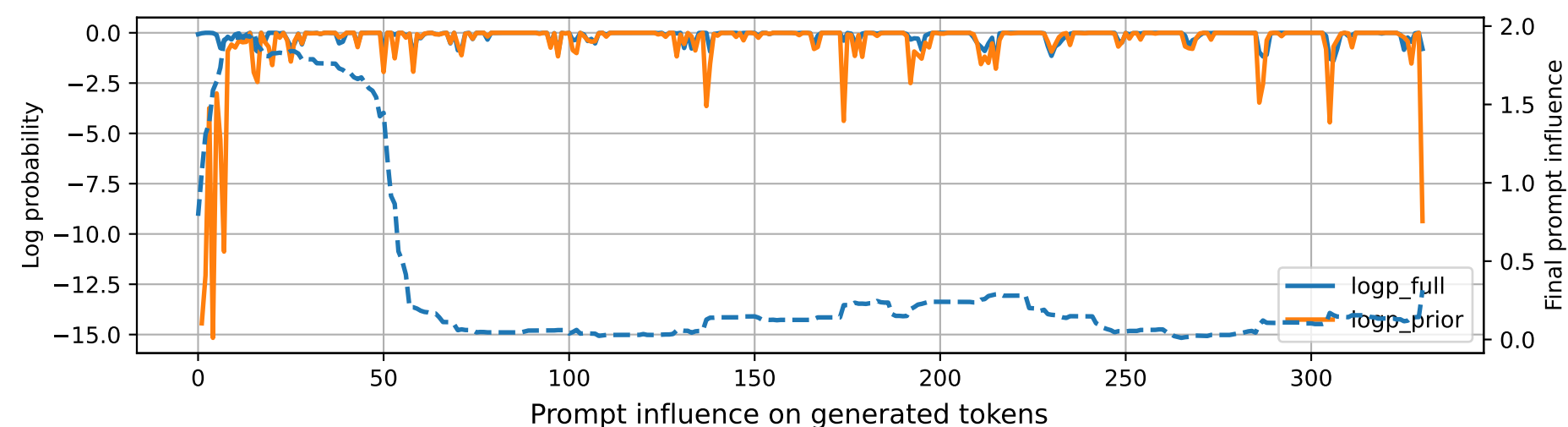


Prompt 1



PROMPT:
Task ID: PYC041 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC041 Your code must contain this required symbol:
def fib Implement this task: define
recursive function fib(n).. Start your answer with the
check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: fib # Recursive function to calculate the nth
Fibonacci number using dynamic programming def
fib(n): # Base case: if n is 0 or 1, return n if n <= 1:
return n # Recursive case:
calculate the Fibonacci number for n-1 and n-2 fib_n_minus_1
= fib(n-1) fib_n_minus_2 = fib(n-2)
Calculate the Fibonacci number for n by summing the two
previous numbers fib_n = fib_n_minus_1 +
fib_n_minus_2 return fib_n # Example usage print(fib(5)) #
Output: 5 print(fib(10)) # Output: 55
print(fib(20)) # Output: 6765 print(fib(30)) # Output: 83205
print(fib(40)) # Output: 115045 print(fib(50))
Output: 161095 print(fib(60)) # Output: 226195 print(fib(70))
Output: 312295 print(fib(80)) # Output:
439395 print(fib(90)) # Output: 618495 print(fib(100)) #
Output: 837595